

Toxic Threat Zone

ALOHA® 5.4.7



Time: July 21, 2025 1721 hours PDT (using computer's clock)

Chemical Name: PROPANE

Wind: 1.55 meters/second from 271° true at 10 meters

THREAT ZONE:

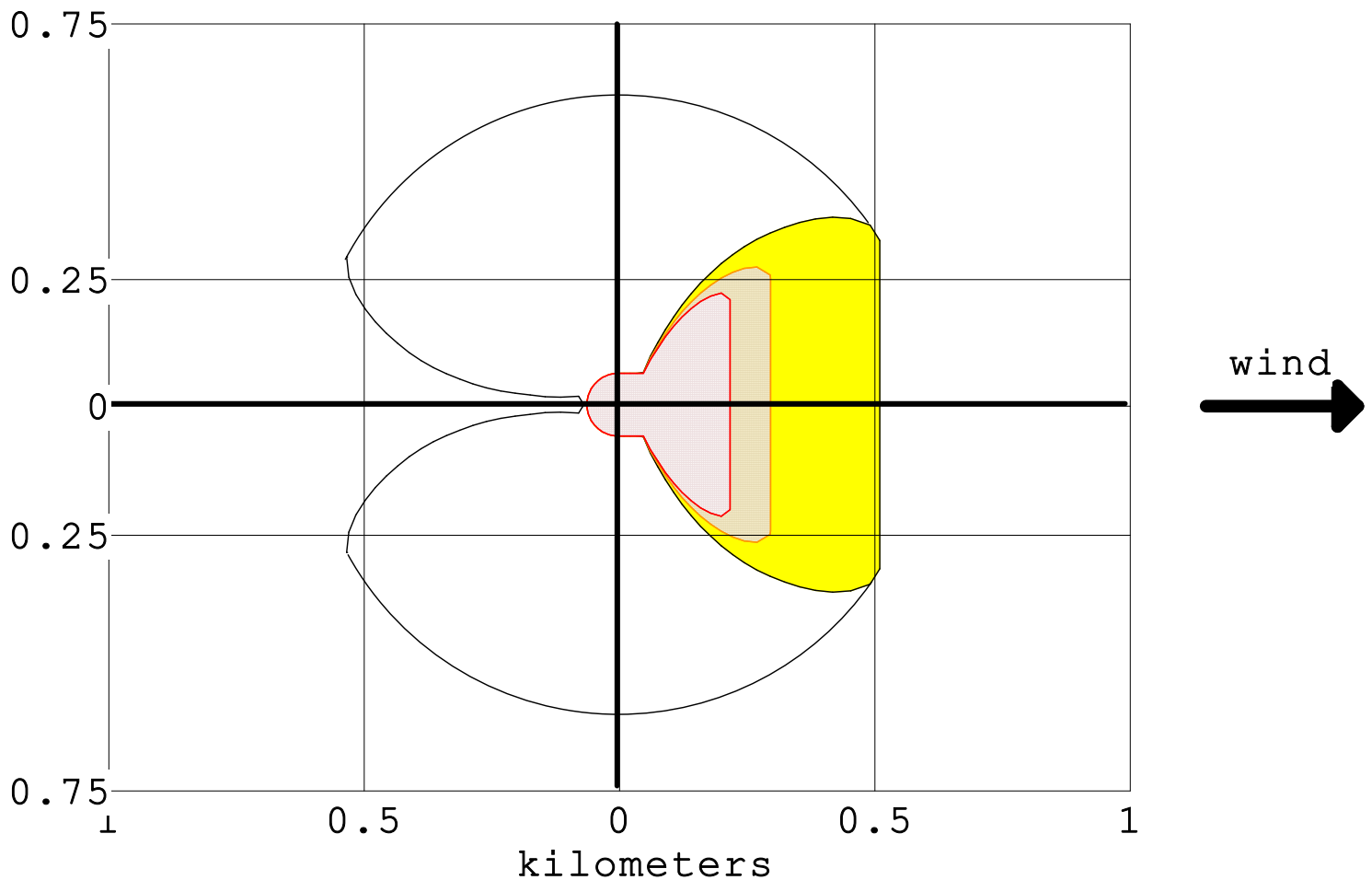
Model Run: Heavy Gas

Red : 220 meters --- (33000 ppm = AEGL-3 [60 min])

Orange: 299 meters --- (17000 ppm = AEGL-2 [60 min])

Yellow: 514 meters --- (5500 ppm = AEGL-1 [60 min])

kilometers



 greater than 33000 ppm (AEGL-3 [60 min])

 greater than 17000 ppm (AEGL-2 [60 min])

 greater than 5500 ppm (AEGL-1 [60 min])

 wind direction confidence lines

Text Summary

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SITE DATA:

Location: TULARE, CALIFORNIA, USA, CALIFORNIA
Building Air Exchanges Per Hour: 0.28 (unsheltered single storied)
Time: July 21, 2025 1721 hours PDT (using computer's clock)

CHEMICAL DATA:

Chemical Name: PROPANE
CAS Number: 74-98-6 Molecular Weight: 44.10 g/mol
AEGL-1 (60 min): 5500 ppm AEGL-2 (60 min): 17000 ppm AEGL-3 (60 min):
33000 ppm
IDLH: 2100 ppm LEL: 21000 ppm UEL: 95000 ppm
Ambient Boiling Point: -42.3° C
Vapor Pressure at Ambient Temperature: greater than 1 atm
Ambient Saturation Concentration: 1,000,000 ppm or 100.0%

ATMOSPHERIC DATA: (MANUAL INPUT OF DATA)

Wind: 1.55 meters/second from 271° true at 10 meters
Ground Roughness: open country Cloud Cover: 4 tenths
Air Temperature: 18.54° C
Stability Class: D (user override)
No Inversion Height Relative Humidity: 54%

SOURCE STRENGTH:

Leak from hole in vertical cylindrical tank
Flammable chemical escaping from tank (not burning)
Tank Diameter: 7 meters Tank Length: 13 meters
Tank Volume: 500 cubic meters
Tank contains liquid Internal Temperature: 18.54° C
Chemical Mass in Tank: 214,418 kilograms
Tank is 85% full
Circular Opening Diameter: 5 centimeters
Opening is 11.0 meters from tank bottom
Release Duration: ALOHA limited the duration to 1 hour
Max Average Sustained Release Rate: 1,910 kilograms/min
(averaged over a minute or more)
Total Amount Released: 91,614 kilograms
Note: The chemical escaped as a mixture of gas and aerosol (two phase flow).

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